Noise Busters

Shot-budget-aware shaded lightcones (SLC) on ibm_kingston

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Final answer: <Z6 Z7> = 0.6788 ± 0.0046, grader score 51.8

Every number reproduces from the job IDs on the last slide.

1. Method: PEC with shaded lightcones, tuned to the shot budget

- Learn a Pauli-Lindblad noise model for the 3 unique layers (NoiseLearnerV3, about 88 s of QPU).
- Shaded lightcones cancel only the noise terms that reach Z6 Z7. Overhead gamma² drops from 7.9x (full PEC) to 3.1x.
- Shot-budget rule: pick the bias tolerance that minimizes sqrt(b² + gamma²/N) instead of cancelling everything and drowning in shot noise.
- Readout: TREX rescaling plus edge post-selection on spectator qubits.

2. Results

Method	Mirror (ideal 1.0)	Run 1 (256 x 128)	Run 2 (512 x 256)
Raw	0.586 ± 0.005	0.5450 ± 0.0046	0.5710 ± 0.0023
TREX + post-selection	0.648 ± 0.005	0.5947 ± 0.0052	0.6199 ± 0.0025
SLC + TREX + post-selection	1.019 ± 0.043	0.6938 ± 0.0102	0.6749 ± 0.0052

- Mirror check passes: the pipeline is wired correctly.
- Final answer = inverse-variance mean of the two SLC edge values: 0.6788 ± 0.0046, score 51.8.
- Submissions 0 to 48.1 to 51.8, each chosen by a rule written before seeing the data. Never compared with the exact value.

3. Honest finding: the limit is bias from a stale noise model, and the queue makes it worse

- Error 0.0165 is 3.5x the statistical error. 4x more shots did not help.
- Raw value drifted 5 sigma in 12 minutes while run 2 reused the 14:12 model. Noise models go stale in about 15 minutes.
- Fix attempted: fresh model plus executor back to back. The job sat in a queue that grew from 8 to 92 jobs and never ran; cancelled at the 4:40 cutoff, nothing new submitted.
- Lesson: queue delay is itself a mitigation bias. Learning and execution must run in one session.

4. Reproduction: job IDs (ibm_kingston, seed_transpiler=1234)

`Chain: [30, 29, 38, 49, 48, 47, 46, 45, 44, 43, 42, 41, 36, 21, 20]`

Job ID	What	Time PT
db5ajpcvf2bc73cvks4g	Baseline Estimator, 10k shots (score 0)	14:11
db5ak64vf2bc73cvkshg	Noise learning, scored circuit	14:12
db5al9qmb58s738a7sa0	Run 1 executor, 256 x 128, tol 1e-3	14:18
db5armklf4us73c3pn20	Run 2 executor, 512 x 256, tol 1e-6	14:30
db5as9clf4us73c3pof0	Mirror noise learning	14:33
db5av2amb58s738a8f70	Mirror executor, 256 x 128	14:39
db5bq4svf2bc73cvmidg	Fresh noise learning, queued, cancelled at cutoff	15:32

Reattach with service.job(id), then select_errors and postprocess from nb_pipeline.py.